

Question	Answer	Marks
6(a)(i)	Any three from: • Light diffracts at the (two) slits. • Light (from each slit) meets / superposes (at the screen). • When the phase difference is 0 (degrees) a bright fringe / (intensity) maximum is formed. • When the phase difference is 180 (degrees) a dark fringe / (intensity) minimum is formed.	B3
6(a)(ii)	$\lambda = ax/D$	C1
	$\lambda = 1.2 \times 10^{-3} \times (10.2 \times 10^{-3} / 6) / 3.1$	C1
	$\lambda = 6.6 \times 10^{-7} \text{ m}$	A1
6(a)(iii)	(new fringe separation will be) smaller	B1
6(b)(i)	A cross at the intersection of the string and the mean position line.	B1
6(b)(ii)	0	A1
6(b)(iii)	180°	A1

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